

Mathematics of Neural Networks

Linear Algebra: Revisiting Linear Regression (STS 311)

You have seen Linear Regression in terms of scalars. But we will rederive linear regression in terms of vectors and Matrices.

Notation:

Scalar

Vectors

Matrices

Handwritten a, b, c
 α, β, γ

a, b, c

A, B, C

Print

a, b, c
 α, β, γ

a, b, c

A, B, C

$\uparrow$ Bolded
small letters

Geometric vector

$\vec{a}, \vec{b}, \vec{c}$

$$\vec{a} = 1\hat{i} + 2\hat{j} + 3\hat{k}$$

$$= 1 \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{\hat{i}} + 2 \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\hat{j}} + 3 \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\hat{k}}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Linear algebra
vector

$$\underline{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \leftarrow \begin{matrix} \text{column} \\ \text{matrix} \end{matrix}$$

3×1

Dot product

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\checkmark \quad \underline{a}^T \underline{b} = \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix}_{1 \times 3} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}_{3 \times 1} \\ = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Transpose of a product

$$(AB)^T = B^T A^T$$

$$(ABC)^T = C^T B^T A^T$$

$$(\underline{a}^T B \underline{c})^T = \underline{c}^T B^T \underline{a}$$

$$(A^T)^T = A$$

Inverse

$$(AB)^{-1} = B^{-1} A^{-1}$$

Common Mistake : Using division with Matrices

$$A \underline{x} = \underline{b} \Rightarrow \boxed{\underline{x} = \frac{\underline{b}}{A}} \times$$

$$A \underline{x} = \underline{b}$$

Left multiply both side with A^{-1}

$$A^{-1} A \underline{x} = A^{-1} \underline{b}$$

$$\Rightarrow \underline{x} = A^{-1} \underline{b}$$

~~$$\underline{x} = \underline{b} A^{-1}$$~~

($\underline{y}$ it exists)

$$\underline{x}^T A = \underline{b}^T$$

Right multiply

$$\underline{x}^T = \underline{b}^T A^{-1}$$

~~$$\underline{x}^T = A^{-1} \underline{b}^T$$~~

$\underline{a}, \underline{b}, \underline{c} \equiv$ column vectors $\begin{bmatrix} \end{bmatrix}_{n \times 1}$

row vector $\begin{bmatrix} \end{bmatrix}_{1 \times n}$
 $\underline{a}^T, \underline{b}^T, \underline{c}^T$

$$\begin{array}{|c|} \hline \underline{a} B \\ \hline \underline{a} \quad \underline{b}^T \\ \hline \underline{a}^T B \\ \hline \end{array} \begin{array}{l} \times \\ \checkmark \\ \checkmark \end{array}$$

$$\begin{bmatrix} \end{bmatrix}_{n \times 1} \begin{bmatrix} \end{bmatrix}_{1 \times m}$$

$$\begin{array}{|c|} \hline \underline{B} \underline{c} \\ \hline \underline{B} \underline{c}^T \\ \hline \underline{b} \underline{c}^T \\ \hline \end{array}$$

✓ ✗ ✓

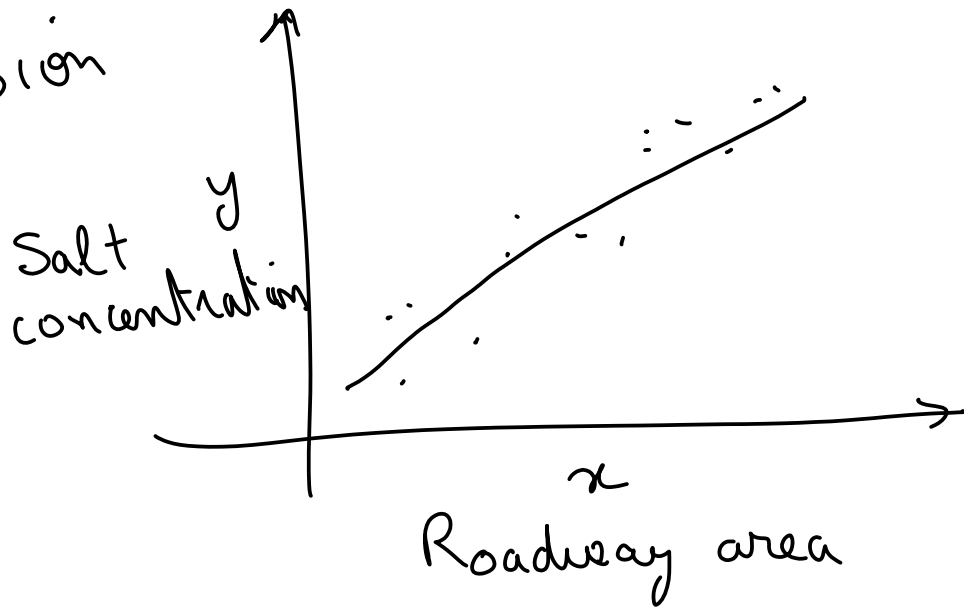
$$\underbrace{\underline{a}^T \underline{X} \underline{B} \underline{y}}_{1 \times n \quad n \times m \quad m \times 1}$$

1 x 1
scalar

Linear Regression

Dataset

$$= \{ (x_1, y_1), (x_2, y_2), \dots, (x_n, y_n) \}$$



Model

$$\hat{y} = \underbrace{\underline{w}^T \underline{x}}_{f(\underline{x})} + w_0$$

Parameters

Loss

$$L(D; w, w_0) = \sum_{i=1}^n \|y_i - \hat{y}_i\|$$

Least squares loss

Training

minimizing w, w_0

$$L(D; w, w_0)$$